Extended reading — Week 25: synthetic ATS RL Enhancement Audit

October 6, 2026

Six papers to read this week

The six papers below are the spine of the audit's argument: counterfactual learning to rank, fair re-ranking in talent search, the LinTS analysis that justifies our Phase-1 algorithm choice, DPO for the LLM components, PPO-Lagrangian for the constrained variant, and the EU AI Act high-risk-system obligations as they bear on recruitment platforms.

1. Joachims, Swaminathan, de Rijke — *Deep Learning with Logged Bandit Feedback* (ICLR 2018, arXiv:1707.07875)

Introduces **BanditNet**, a counterfactual-risk-minimisation estimator that lets neural networks be trained directly from logged bandit feedback without an interaction loop. The key insight is the self-normalised IPS objective with a variance-regularisation term.

Why it matters this week. The Phase-1 reranker is exactly this setting — we have logged ranking decisions and propensities, we want to train a model offline without re-running the recruiter loop. The SNIPS variant we use in `project/ope.py` is the same estimator BanditNet builds its loss around.

> "We propose Counterfactual Risk Minimization (CRM) for training deep networks using batch logged bandit feedback. Our experiments show that the proposed CRM objective with the self-normalised estimator and a variance regulariser significantly outperforms approaches that maximise predictive likelihood or use the naive IPS estimator."

Commentary. The variance regularisation is the practical bit. A pure SNIPS objective will happily converge to a deterministic policy even on tiny data, with absurd importance weights. The CRM regulariser penalises high-variance solutions and is the reason BanditNet works out-of-the-box on real logs. Carry this into any production OPE work.

2. Geyik, Ambler, Kenthapadi — *Fairness-Aware Ranking in Search & Recommendation Systems with Application to LinkedIn Talent Search* (KDD 2019, arXiv:1905.01989)

LinkedIn's deployed post-processing reranker that enforces a target demographic distribution on the top-k. Three algorithms (DetGreedy, DetCons, DetRelaxed) trading utility against constraint tightness; the authors deployed DetGreedy at scale.

Why it matters this week. This is the closest published production reference for what we propose at D1 + the CMDP constraint. The audit borrows two of their tricks verbatim: (i)

measure parity at the top-k slate, not at the full ranking; (ii) treat the fairness-utility trade-off as a frontier, not a single point.

> "We propose a measurement framework for assessing fairness of ranked results, [...] and a re-ranking framework that ensures fairness while having no significant impact on business metrics."

Commentary. The "no significant impact on business metrics" clause is the policy-deployment language we need for the Phase-2 go/no-go gate. Without an explicit business-metric guarantee, fairness work loses internal champions fast. Bake the dual evaluation into the roadmap.

3. Agrawal & Goyal — *Thompson Sampling for Contextual Bandits with Linear Payoffs* (ICML 2013, arXiv:1209.3352)

Frequentist regret analysis of LinTS. Bounds the probability that a sub-optimal arm's posterior dominates the optimal arm's, and yields $\tilde{O}(d^{3/2}\sqrt{T})$ regret.

Why it matters this week. The Phase-1 algorithm. The $d^{3/2}$ scaling matters because the candidate-job feature vector is $d = 16$ in our demo and would be $d \approx 60{-}100$ in production (skill embeddings + agent scores + confidence). LinTS scales gracefully; LinUCB needs more care with the confidence ellipsoid radius at large d.

> "At each time step, the algorithm picks an action by drawing a sample from the current posterior distribution over the parameter and choosing the action that maximises the expected reward given that sample."

Commentary. The deceptively simple sample-then-greedy mechanism has two industrial virtues that drove its adoption at Yahoo! and Microsoft: (a) it parallelises trivially (every shard samples its own $\tilde{\theta}$); (b) it remains well-calibrated even when the posterior is approximate (e.g. Laplace approximation), unlike UCB whose confidence ellipsoid degrades sharply.

4. Rafailov, Sharma, Mitchell, Manning, Ermon, Finn — *Direct Preference Optimization: Your Language Model is Secretly a Reward Model* (NeurIPS 2023, arXiv:2305.18290)

DPO. The paper that replaced RLHF's two-step (reward model → PPO) with a single closed-form preference loss. Eliminates the reward-model overfitting failure mode and is far cheaper to train.

Why it matters this week. D5 (outreach copy) and D11 (JD polishing) are exactly the regime DPO was designed for: we have logged recruiter preferences over pairs of LLM drafts, we want to update the draft model to align with those preferences, and we cannot afford a full RLHF stack. DPO is the Phase-4 recipe.

> "Our experiments show that DPO is at least as good or better than existing methods, including PPO-based RLHF, while being substantially simpler to implement and train."

Commentary. The under-discussed risk with DPO is length bias (week 19, week 21): the closed-form solution can drift to longer responses regardless of preference content. The mitigation (length-controlled DPO; Park et al. 2024, arXiv:2403.19159) is the single must-cite extension when arguing the production case.

5. Stooke, Achiam, Abbeel — *Responsive Safety in Reinforcement Learning by PID Lagrangian Methods* (ICML 2020, arXiv:2007.03964)

PID-controlled Lagrange multiplier for CMDPs — the practical recipe that makes PPO-Lagrangian actually work. Plain Lagrangian methods oscillate; PID damping converges.

Why it matters this week. The Phase-3 constrained learner is a PPO-Lagrangian. The cost in our case is a fairness metric; the PID machinery is what keeps the multiplier from oscillating around the constraint boundary as the applicant pool drifts.

> "We treat the Lagrange multiplier update as a control problem and apply PID control to it, achieving improved stability and final performance over standard Lagrangian methods."

Commentary. The PID gains are the new hyperparameters you have to tune. Stooke et al. provide a small grid; we recommend starting with their reported values for the Safety Gym suite and re-tuning against the simulated parity drift in our `constrained.py` demo.

6. Future of Life Institute (eds.) — *The EU AI Act: A Practical Guide for Recruitment AI* (FLI policy brief, 2024–2025)

Walks through Annex III point 4 (recruitment), Articles 9 (risk management), 13–15 (transparency / accuracy / oversight), and 26 (deployer obligations) as they apply concretely to ATS platforms. Recommends operating models, audit cadences, and documentation templates.

Why it matters this week. Chapter 7 of the notes leans on the Act. The brief is the most concise compliance-grade reference for defending the audit's roadmap to a legal review board.

> "Providers of high-risk AI systems used for recruitment or selection of natural persons must implement a quality management system, maintain technical documentation, log decision events with sufficient detail to support audits, and ensure that affected persons can request human review."

Commentary. The Act does not prescribe an algorithm. It prescribes a process. The audit's Phase 0 (instrument logging, introduce ε-soft exploration) is the cheapest, fastest route to Act compliance — without it, no later phase can be defensibly deployed in the EU. Even teams outside the EU should treat this as the de-facto standard; UK and US regulators (ICO, EEOC) are converging on similar requirements.

Sourced from `research/bandits_matching_sota.md`, `research/safety_rl_sota.md`, and `research/llm_rl_sota.md` in this repository.